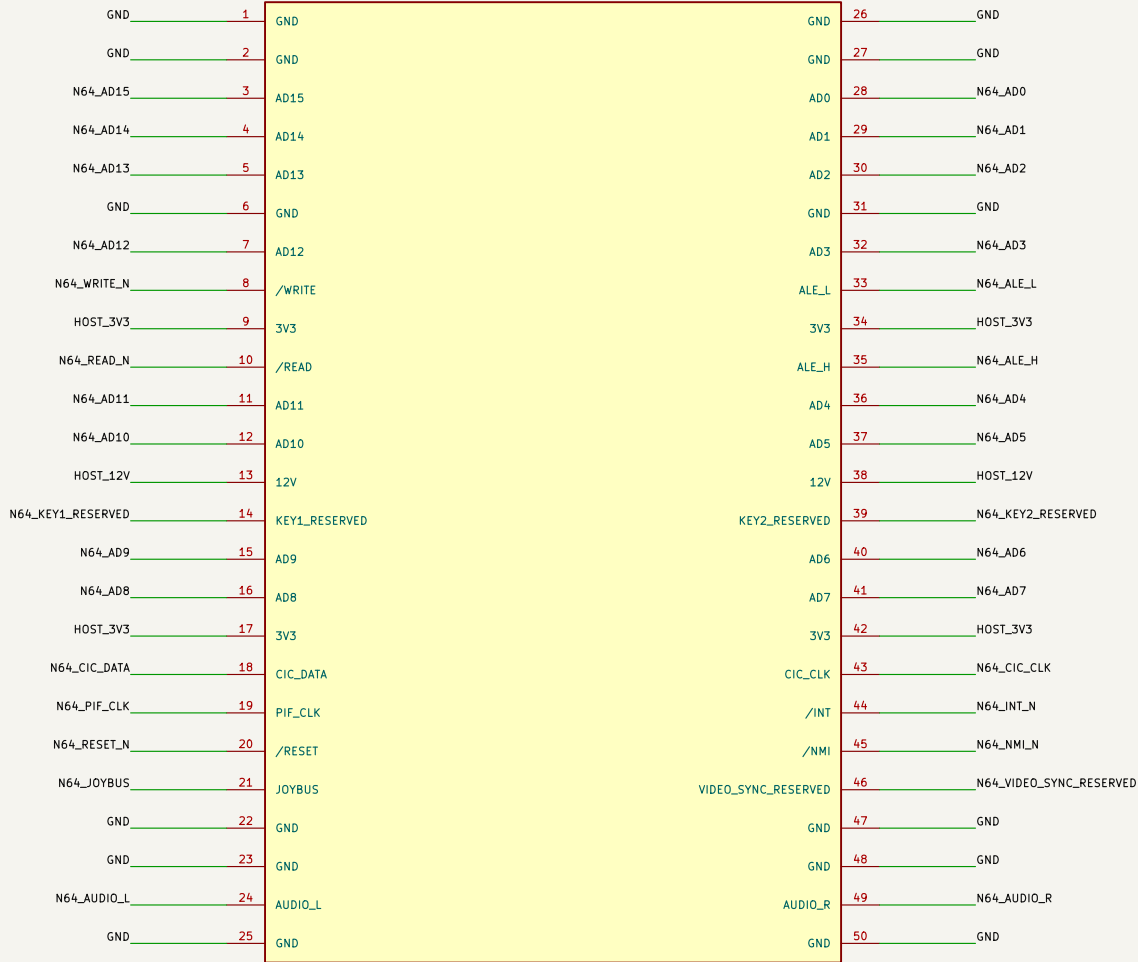


N64 / M64 HOST EDGE

Same 50-contact interface for both host systems

J1
N64_Cartridge_Edge_50



HOST INTERFACE
3.3 V logic; power inputs remain separate from local rails.
AD[15:0] is bidirectional. CIC_DATA / JOYBUS require release.
Reserved contacts and audio are identified only; no drivers fitted.
HOST_12V is reserved, with no selected load. No 5 V edge supply.

USB-C programmer

GND

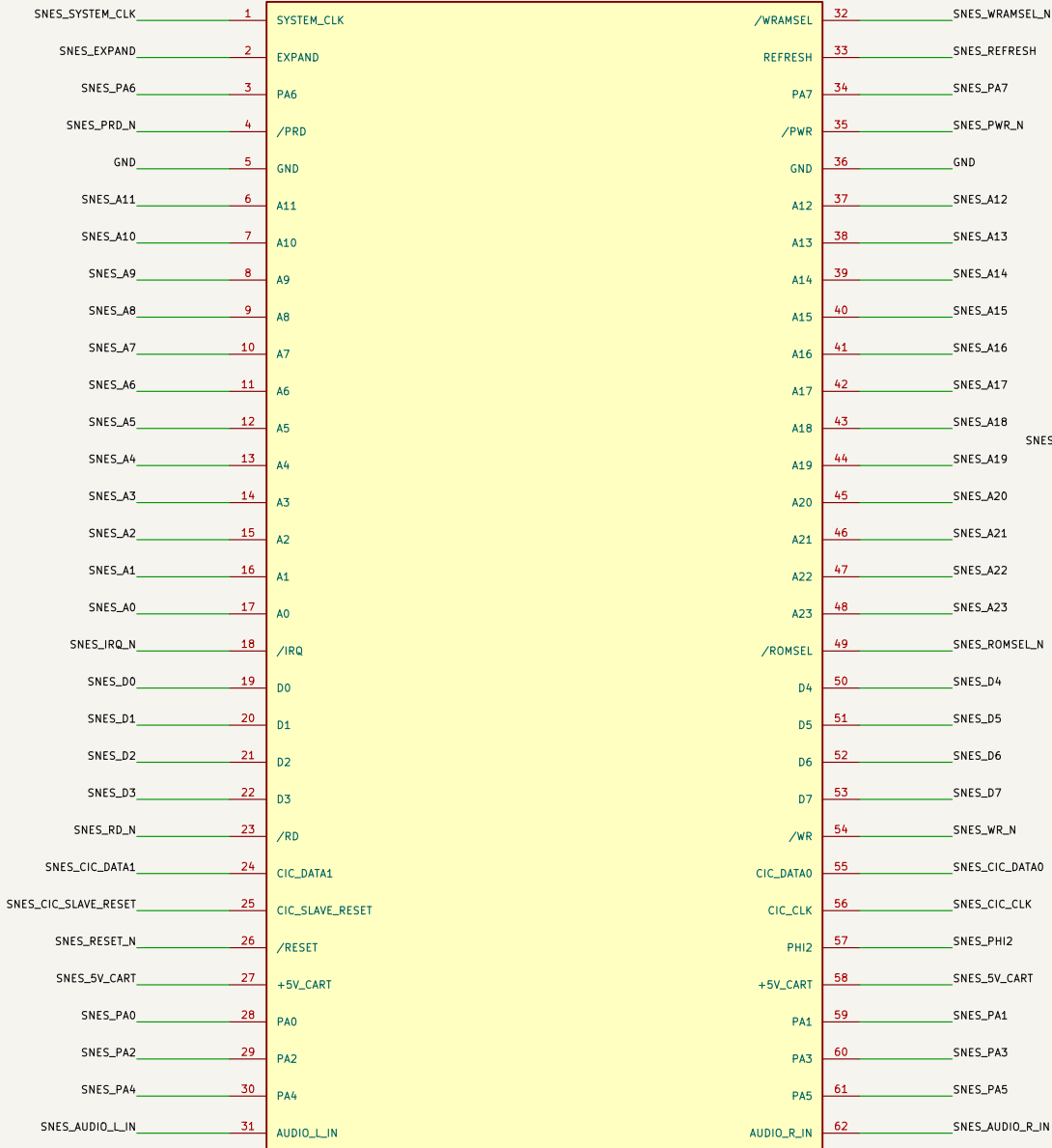
File: usb-programmer.kicad_sch

DRAFT BOUNDARY: cartridge connectors + USB programmer child sheet. Main FPGA, system power, cartridge translation and A/V remain pending.
Only GND is common between the two connectors. Matching names within one domain identify the same net; _N means active low.
Connector pin types are passive; electrical directions and source evidence are in interfaces/*_csv and docs/design/*-interface-notes.md.

SNES / SUPER FAMICOM SOCKET

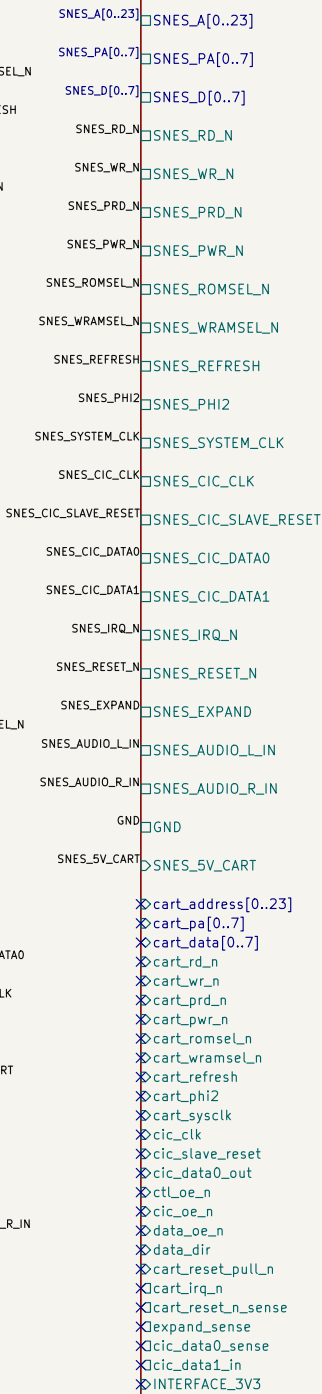
All 62 contacts retained, including analog cartridge audio

J2
SNES_Female_Slot_62



SNES INTERFACE
5 V cartridge logic requires reviewed translation and isolation.
SNES_5V_CART: protected switched supply, default OFF (pending).
System RESET must support cartridge assertion and input sensing.
CIC_SLAVE_RESET is separate; AUDIO_L/R_IN are analog inputs.
No universal NC contacts. Full pin behavior is recorded in the CSV.

SNES cartridge interface



File: cart-interface.kicad_sch

Source attribution and limits: hardware/sn64/README.md
INTERFACE DRAFT ONLY – NOT FOR MANUFACTURE

SN 64

Sheet: /
File: sn64.kicad_sch

Title: SN 64 – cartridge interfaces

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-28

Rev: 0.2-usb

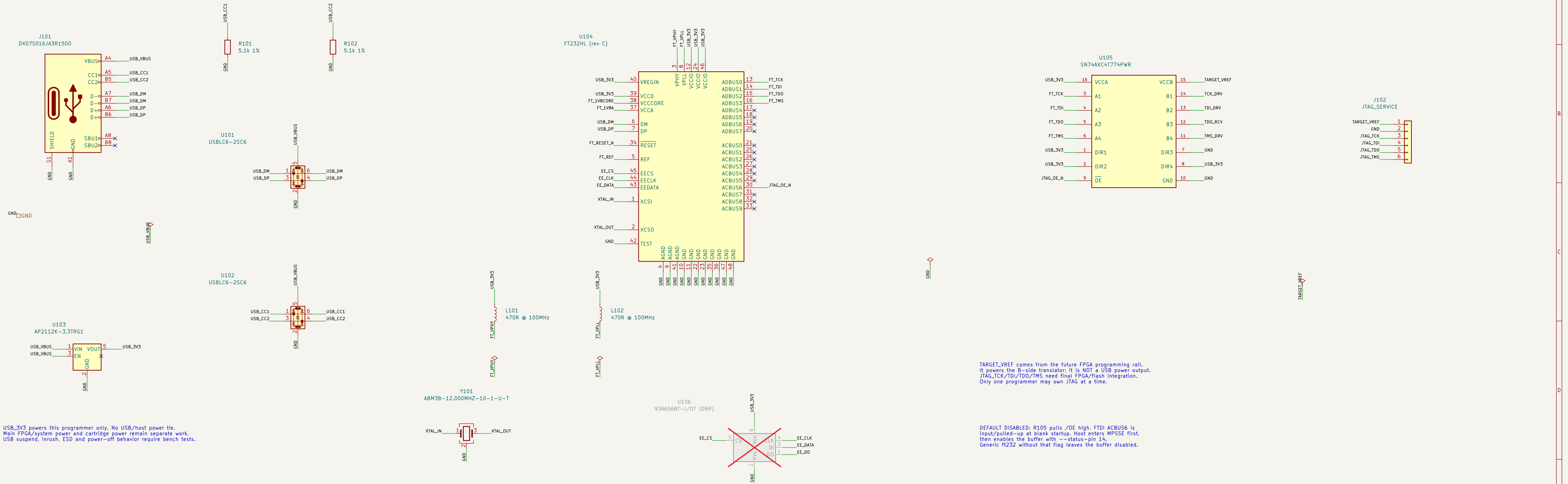
Id: 1/3

SIDE USB-C: INITIAL PROGRAMMING + RECOVERY

USB 2.0 device: 5V input only. Side location is mandatory: exact PCB/enclosure placement remains open.

FT232H HARDWARE BRIDGE

TARGET ISOLATION + SERVICE ACCESS

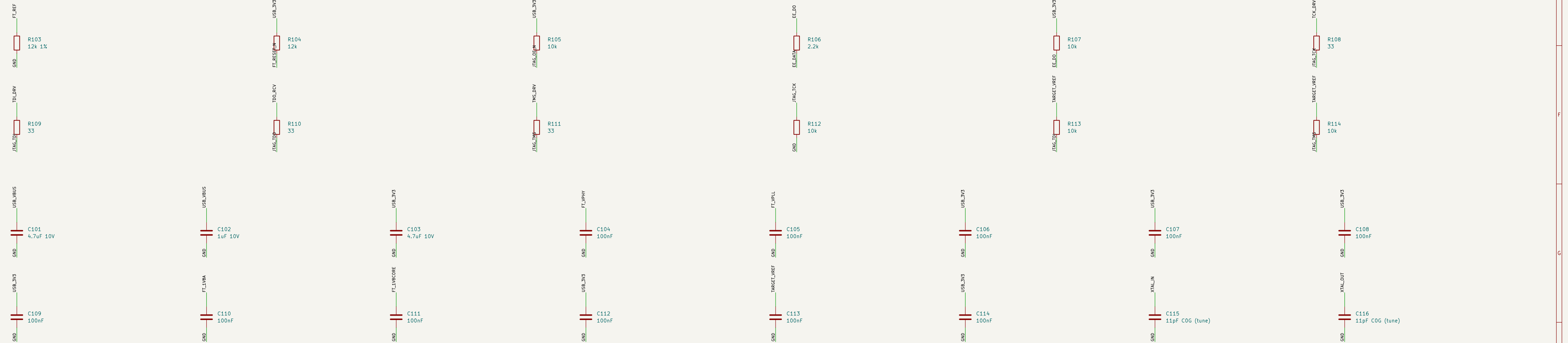


USB_3V3 powers this programmer only. No USB/host power tie. Main FPGA/system power and cartridge power remain separate work. USB suspend, inrush, ESD and power-off behavior require bench tests.

TARGET_VREF comes from the future FPGA programming rail. It powers the 8-side translator; it is NOT a USB power output. JTAG_TCK/TDI/TDO/TMS need final FPGA/flash integration. Only one programmer may own JTAG at a time.

DEFAULT DISABLED: R105 pulls /OE high. FTDI ACBUS6 is input/pulled-up at blank startup. Host enters MPSSE first, then enables the buffer with --status-pin 14. Generic ft232 without that flag leaves the buffer disabled.

SUPPORT PASSIVES (place at their associated IC/pin; electrical grouping here is not PCB placement)



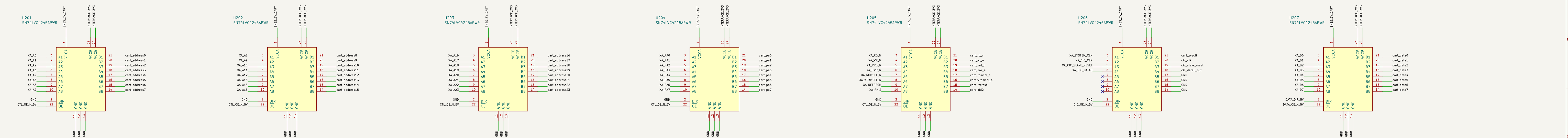
DRAFT 0.2: USB programmer circuitry only. FPGA/storage choice, persistent-image map, system power, routing and physical validation are pending. Sources: SummerCart64 a1e7996 (CERN-DHL-S-2.0), FTDI FT232H v2.2, TI SN74AXC4774, JAE drawing SJ122205. See docs/design/usb-programming-architecture.md.

SNES CARTRIDGE – SIDE TRANSLATION AND PROTECTION

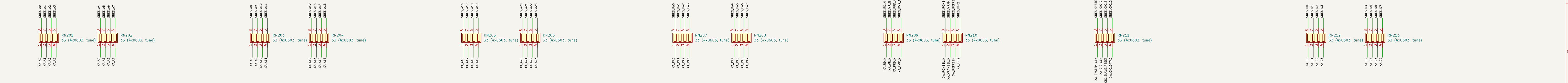
SN74LVC245A: A port = 5 V VCCA = SNES_SV_CART; B port = 3.3 V VPCB = INTERFACE_3V3; DIR/DE are referenced to VCCA (TI SCAS375K). DIR=H = A<B; OE=H: isolation.

U201–U206: console outputs, DIR strapped to GND (fixed FPGA→cartridge). U201–U205 share CTL_DE_N_SV (bridge ctl_de_n); U206 (SYSTEM_CLK + CIC_CLK/SLAVE_RESET/DATA0) has its own CIC_DE_N_SV (proposed bridge port ctc_de_n).

U207: DS-BY, DIR and OE both FPGA-controlled (bridge data_dir / data_de_n). Unused B inputs tied to GND; unused A outputs no-connect.



CARTRIDGE OUTPUT DAMPING: 33 Ω arrays at the translator A pins (source-series). Values are provisional; tune after SI measurement.



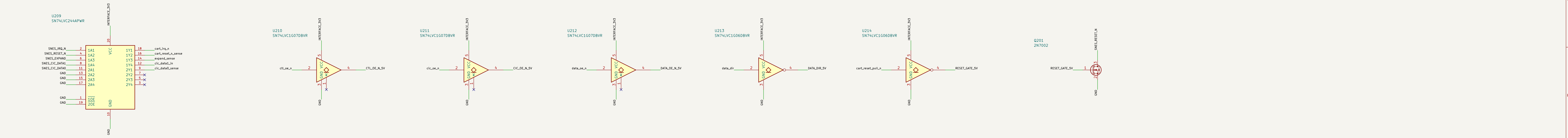
SENSE + RELEASE PATHS

U209 SN74LVC244A at 3.3 V, 5.5 V-tolerant inputs with buff. recovers /IRQ, /RESET, EXPAND, CIC_DATA1 and CIC_DATA0 (sense while released).

U210–U212 74LVC1607 / U213 74LVC1606: 3.3 V open-drain outputs (5.5 V tolerant, I_{OFF} pulled to SNES_SV_CART).

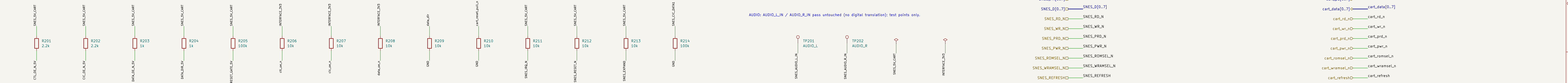
No 5 V pull-up reaches an FPGA pin; cartridge pull-off or FPGA unclocked → every transition OE released high (isolated).

/RESET: Q201 open-drain sda. RESET_GATE_SV pulled to 5 V; cartridge held in reset unless INTERFACE_3V3 is up AND the FPGA drives cart_reset_pull_n=1.

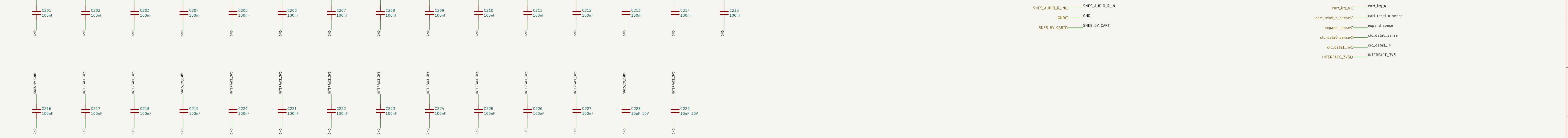


/OE and DIR edge timing: the bridge releases the data octet one A6.56 ns clock before turnaround. R203/R204 = 1k keep the passive rising edge short (about 10 pF load; tau < 10 ns). R201/R202 2.2k drive fty/one /OE inputs where only permit-loss timing matters. Verify on hardware.

PULL/BIAS: /IRQ, /RESET, EXPAND 10k to cartridge 5 V (OpenSFC keeps pull-ups on IRQ/EXPAND); CIC_DATA1 100k idle bias. Values provisional.



DECOUPLING: only 100 nF per supply pin; interface at the pin) plus one 10 μF bulk per rail.



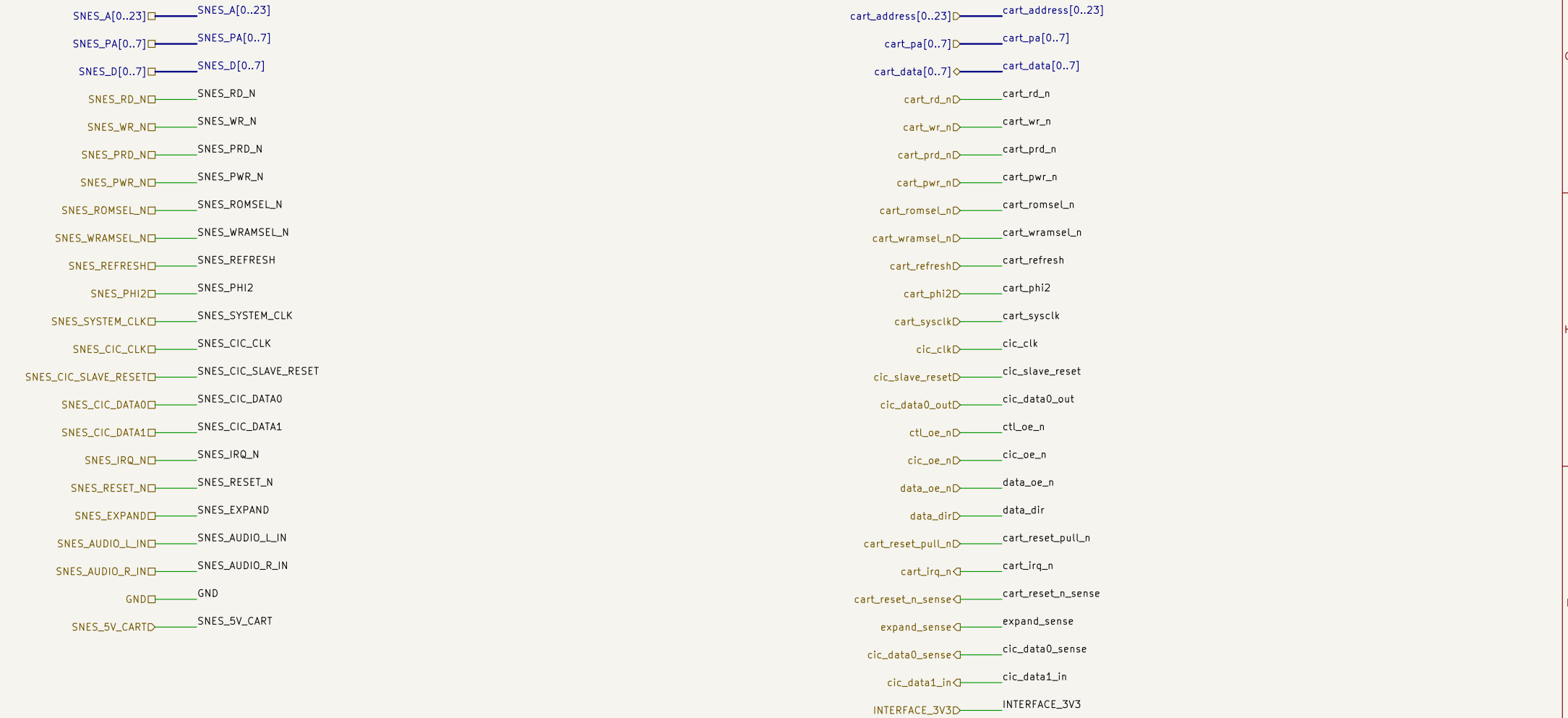
DBMT 0.3-cart schematic circuit candidate only. No PCB. SI simulation, power sequencing tests, ESD part or cartridge test has been performed.

SNES_SV_CART and INTERFACE_3V3 are hierarchical inputs from the (future) power sheet: #U201/#U202 mark them externally supplied for ERC only.

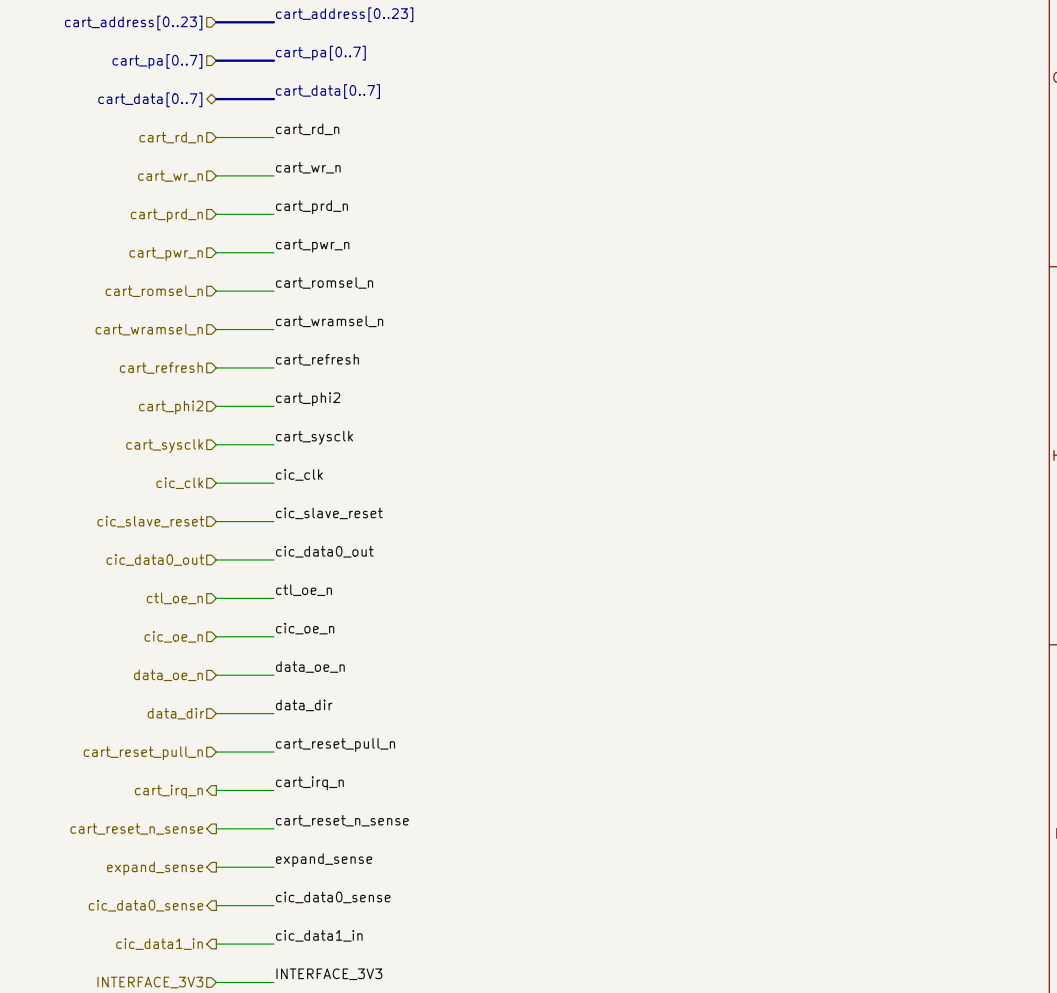
Open: select ESD array selection; independent CIC_DATA0 release; SuperCIC pull-mode drive in CIC_DATA1; final damping/pull values.

Sources: TI SCAS375K (SN74LVC245A), SCAS414M6 (SN74LVC244A), SCE5296A6 (SN74LVC1607), SCE5296A8 (SN74LVC1606). See docs/design/cart-interface-schematic.md.

HIERARCHICAL PORTS – CARTRIDGE SIDE (root J2 net labels)



PENDING PORTS – FPGA SIDE (bridge port names) + DEFAULT-OFF RAILS FROM FUTURE POWER SHEET



DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Shiren / SNES cartridge interface /

File: cart-interface.kicad_sch

Title: SN 64 – SNES cartridge interface

Size: A1 Date: 2026-09-29

Kicad E.D.A. 10.0.6

Rev: 0.3-cart
1 of 3/3